

A Multimodal Fusion Framework for Solar Forecasting using Dynamic GNNs and Mixture of Experts

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Abstract

Accurate solar PV forecasting across minutes-to-day horizons is difficult due to multi-scale cloud dynamics, non-stationary irradiance, and heterogeneous data sources. We introduce a multimodal deep-learning framework that fuses sky imagery, satellite patches, ground irradiance/meteorological time series, and mesoscale model fields from nearby grid nodes. The model (i) predicts clear-sky-index residuals to reduce diurnal/seasonal non-stationarity, (ii) performs an adaptive fusion that dynamically readjusts the importance of each data source (e.g., sky images, satellite) on a per-sample basis at inference time, (iii) uses a spatial GNN with meteorology-driven dynamic adjacency to share context across stations, and (iv) employs a mixture-of-experts head with load-balanced, differentiable routing to jointly produce all horizons.

Evaluated on a comprehensive multi-year dataset, our framework consistently outperforms a suite of state-of-the-art deep learning baselines, including various Transformer architectures (PatchTST, Crossformer, Informer) and Temporal Convolutional Networks (TCN). For challenging day-ahead forecasting, our model reduces the average Root Mean Square Error by nearly 15% compared to the next-best Transformer model, achieving a forecast skill of over 49% relative to the NAM physical model, especially under ‘rampy’ (high variability) weather conditions. This advantage is maintained across shorter time scales, demonstrating a 9.5% error reduction on intra-day horizons and a clear performance lead for intra-hour forecasts. These results highlight the framework's ability to leverage complementary data sources through its novel fusion and routing mechanisms, establishing a new state-of-the-art for robust and accurate forecasts across diverse weather conditions and time horizons.

Keywords: PV forecasting, deep multimodal fusion, sky imagery, satellite images, graph neural networks (GNN), Mixture of Experts (MoE).

1. Introduction and Limitations of Existing Studies

The accelerating global adoption of solar photovoltaic (PV) technology, driven by carbon-neutrality targets, has made accurate PV power forecasting a cornerstone of modern grid operations. Solar generation is inherently variable due to multi-scale atmospheric processes, ranging from rapidly evolving local cumulus clouds to synoptic-scale systems. These variations can cause abrupt ramps in PV output, posing risks to grid stability and increasing reliance on costly reserves. Reliable short- and long-term forecasts enable operators to balance supply and demand, optimize energy storage dispatch, and reduce curtailment, thereby improving both economic efficiency and renewable penetration [1].

A range of methodologies has been explored for PV forecasting, each with distinct strengths and limitations. Physics-based approaches [2], [3], including numerical weather prediction (NWP) models, simulate atmospheric dynamics and radiative transfer with physically consistent outputs. However, they require high-performance computing resources and have substantial latency, making them less suitable for intra-hour forecasting. Their multi-kilometer grid resolutions cannot resolve sub-kilometer cloud structures, and their reliance on complex data assimilation can distort high-frequency observational features critical for short-term accuracy [4].

Statistical models such as autoregressive integrated moving average (ARIMA), vector autoregression (VAR) and linear regression are widely used in short-term solar forecasting because they are computationally light and easy to implement. These models rely on historical irradiance data and often use engineered features like moving averages or clearness indices to capture trends. However, their linear structure makes them ill-suited for capturing the nonlinear interactions between cloud dynamics, atmospheric conditions and solar output; they perform poorly during extreme or uncharacteristic weather events [5]. Classical machine-learning methods (e.g., support vector regression, random forests) can model some nonlinearities but still require extensive feature engineering and hyper-parameter tuning, and they struggle to adapt when weather regimes shift or novel patterns emerge [6]. Furthermore, these models typically treat each site independently, ignoring spatial correlations that are crucial for multi-site forecasting. Even when hybridized with NWP outputs, statistical corrections often provide only modest improvements, highlighting the need for more expressive models [7].

Deep learning approaches have advanced the state of the art by learning nonlinear spatiotemporal representations directly from high-dimensional data. Convolutional neural networks (CNNs) and ConvLSTMs leverage sky or satellite imagery to model cloud evolution [8], while recurrent and temporal convolutional networks (LSTMs, TCNs) capture sequential dependencies in irradiance and meteorological time series [9]. Graph neural networks (GNNs) have been introduced to model spatial dependencies between sites [10]. However, most existing deep learning models remain single-modality—image-based models cannot incorporate ground measurements, and time-series models lack visual context [11], [12]. Moreover, fixed-graph GNNs assume static spatial relationships, limiting adaptability to transient meteorological patterns. These constraints, combined with interpretability and generalization challenges, highlight the need for more flexible, adaptive architectures.

The importance of multimodal fusion arises from the complementary strengths of different data sources. By integrating diverse modalities, fusion techniques have advanced a wide range of fields [13], including robotics (sensor integration for robust perception) [14], medical

imaging (combining scans and clinical data for improved diagnosis) and signal processing [15], navigation and localization (visual–LiDAR–IMU fusion for reliable positioning) [16], and video encoding [17]. Sky images deliver high-temporal-resolution observations of local cloud fields, while satellite imagery captures mesoscale cloud systems [18]. Ground-based irradiance and meteorological sensors record point-level atmospheric states, and mesoscale model fields, such as those from the North American Mesoscale Model (NAM), provide forecasts of large-scale atmospheric evolution. Single-modality models fail to leverage these complementarities, reducing robustness under diverse conditions. Adaptive multimodal fusion enables a forecasting model to dynamically prioritize the most informative modalities per sample and forecast horizon, improving both accuracy and stability [19].

Graph-based spatial modeling further enhances predictive skill by propagating contextual information between geographically distributed PV sites. Spatiotemporal GNNs extend this capability by integrating temporal dynamics, but the widespread use of static adjacency matrices—based solely on geographic proximity—limits their responsiveness to evolving weather conditions [20]. Empirical evidence suggests that such static graphs can incur significantly higher error in high-variability regimes, underscoring the need for dynamic, meteorology-aware connectivity.

Our proposed framework directly addresses these gaps with a set of integrated solutions. To overcome the limitations of static graphs, we introduce a meteorology-driven dynamic adjacency mechanism, computing cosine similarity between normalized mesoscale (NAM) features at each time step to adaptively determine inter-site influences. To solve the single modality-problem, this is combined with an adaptive multimodal fusion block that reweights sky image, satellite, time-series, and spatial-graph features on a per-sample basis. Furthermore, a horizon-aware mixture-of-experts head with differentiable Gumbel–Softmax gating ensures forecast specialization without training separate models. Finally, by predicting clear-sky-index residuals, the model directly mitigates diurnal and seasonal non-stationarity that challenges traditional models. The core innovation lies not in a simple combination of these components, but in their organic integration: the meteorology-driven dynamic GNN and the sample-wise adaptive fusion create a dual-adaptive system specifically tailored to solar forecasting. Together, these design choices overcome the resolution, generalization, and single-modality limitations of prior work, delivering consistent accuracy improvements across minutes-to-day horizons.

2. Proposed System Architecture

We propose a multimodal deep learning framework for solar photovoltaic (PV) forecasting that is designed to leverage the complementary strengths of different data modalities, enabling more accurate and robust forecasts. By combining local and mesoscale information, the model addresses the limitations of single-modality approaches, such as the incomplete capture of atmospheric dynamics. The overall architecture, illustrated in Fig. 1 integrates four distinct input modalities using specialized encoder backbones. Dual Vision Transformers (ViT) encode sky and satellite imagery, a Temporal Convolutional Network (TCN) captures patterns in surface weather and irradiance time series, and a Spatio-Temporal Graph Neural Network (GNN) learns representations from the gridded NAM data. These parallel embeddings are then integrated by a central Adaptive Fusion module, which uses a lightweight gating network to learn fusion weights from the concatenated embeddings, creating a unified multimodal representation. Finally, this rich embedding ($\mathbf{z}_t$) is passed to a Horizon-Aware MoE Decoder to generate robust and accurate GHI predictions from short to long hours.

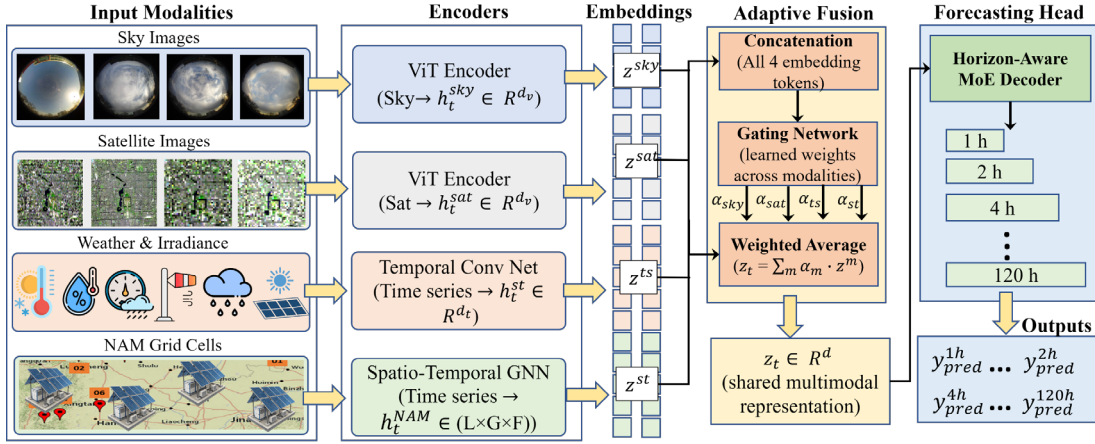


Fig. 1. The architecture of the proposed multimodal fusion model. Four input modalities are processed by specialized encoders. The resulting embeddings are integrated by an adaptive fusion module, and the final representation is passed to a multi-horizon forecasting head to predict GHI from short to long hours ahead.

2.1 Input Modalities and Preprocessing

Notation. We use t for the current decision timestamp on the 5-min grid and L for the look-back window length. The forecast horizon index is $h \in \{1, \dots, H\}$ with H total lead times. $\mathbb{R}$ denotes the real domain, and bold uppercase/lowercase symbols (e.g., X, z) denote tensors/vectors. The operator $\text{clip}(x, a, b)$ clamps x to $[a, b]$. i denotes an identity matrix. Unless stated otherwise, all z-scoring statistics are fit on the training split and applied to validation/test.

2.1.1 Input Modalities

To capture a comprehensive view of atmospheric conditions affecting solar irradiance, we utilize four complementary data modalities. Each modality provides unique insights: high-resolution local details, broader regional context, ground-level measurements, and simulated weather patterns. Below, we describe each modality and its preprocessing steps to ensure data consistency and model compatibility.

- **Sky Images.** These are sequences of fisheye sky-camera images that capture the evolution of local cloud cover over time. Sky images provide high temporal resolution, making them particularly effective for detecting rapid changes in cloud motion, density, and shading patterns near the PV site. We represent the sky image sequence as (1):

$$X^{sky} \in \mathbb{R}^{H \times W \times 3 \times T} \quad (1)$$

where H and W represent the image height and width in pixels, and 3 is for RGB color channels. The number of consecutive frames in the temporal sequence is denoted as T . Thus, X^{sky} encodes both spatial cloud structures (per frame) and temporal cloud motion (across frames), crucial for short-term PV ramp forecasting.

- **Satellite Imagery.** Satellite imagery captures large-scale cloud fields and mesoscale weather systems beyond the PV site's immediate field of view. These inputs provide regional context that complement localized sky-camera observations, enabling forecasts at longer horizons (hours to day-ahead). We represent satellite data as (2):

$$X^{sat} \in \mathbb{R}^{H_s \times W_s \times C_s \times T} \quad (2)$$

where H_s and W_s are spatial resolution of the satellite grid, C_s is number of spectral channels (e.g., visible, infrared, water vapor), and T is temporal sequence length. Satellite imagery captures macro-level cloud advection and weather fronts, bridging the gap between local sensors and global forecasts.

- **Ground Weather and PV Time Series.** Ground-based measurements provide direct observations of irradiance, temperature, wind speed, humidity, and PV output at the site level. These features reflect real-time atmospheric conditions and PV system response, offering strong predictive signals for both short- and medium-term horizons. The time-series modality is represented as (3):

$$X^{ts} \in R^{F \times T} \quad (3)$$

Where, F is number of features including GHI, DNI, temperature, humidity, wind speed, PV output. T denotes temporal sequence length.

- **Meteorological Fields.** Numerical weather prediction (NWP) reanalysis or mesoscale models such as the North American Mesoscale (NAM) system provide structured meteorological fields (e.g., pressure, wind vectors, temperature, cloud fraction) on a regional grid. Unlike satellite imagery, these fields are physically simulated and extend to variables that are not directly observable (e.g., geopotential height, vorticity). We represent the NAM features as (4):

$$X^{NAM} \in R^{G \times F_n \times T} \quad (4)$$

where G is the number of grid points (stations) around the PV site, F_n is the number of meteorological variables (e.g., wind-u, wind-v, temperature, humidity, pressure), and T is the temporal sequence length.

Meteorological fields are particularly valuable for learning dynamic spatial dependencies among nearby nodes. In our framework, they are further used to compute a time-varying adjacency matrix for graph learning. Let $X_t^{NAM} \in R^{G \times F_n}$ denote the NAM slice at time t (i.e., the t -th frame of Eq.(4)). We L2-normalize node feature vectors row-wise to obtain $\tilde{X}_t^{NAM}$. We then compute a dense similarity matrix and sparsify it to form the dynamic adjacency in (5) and (6):

$$S_t = \tilde{X}_t^{NAM} (\tilde{X}_t^{NAM})^T \quad (5)$$

$$A_t = \text{TopKRow}(S_t + \beta I, k) \quad (6)$$

Here, $\tilde{X}_t^{NAM}$ denotes row-wise L2-normalized NAM features, $I \in \mathbb{R}^{G \times G}$ is the identity matrix, and β is a scalar self-bias added to diagonal entries to stabilize self-influence. Whereas, $S_t(i, j) = \cos(x_{t,i}^{NAM}, x_{t,j}^{NAM})$ is the cosine similarity between the normalized NAM states of nodes i and j ; β is a self-consistency bias that stabilizes diagonal (self) influence when cross-node correlations are weak; and $\text{TopKRow}(\cdot, k)$ retains the top- k neighbors per node. Because A_t is computed solely from inputs available at time t , this construction uses *no future information (no lookahead)* and is updated at every time step to reflect evolving meteorological regimes.

Efficient implementation. Although, (5) implies per-step re-computation, the overhead is minimal due to tensor vectorization. Specifically, S_t is obtained by a single (batch) matrix multiplication of size $(G \times F_n) \cdot (F_n \times G)$, producing a $(G \times G)$ similarity matrix. With the small number of regional NAM nodes used in this study ($G = 4$), the adjacency computation is negligible compared to the ViT-based feature extraction overhead in the vision branches, preserving real-time inference (approximately 5.91 ms per sample) even with dynamic graph updates.

2.1.2 Preprocessing

All modalities are synchronized to a common 5-min grid over a fixed look-back window, with sky images, satellite patches, ground irradiance/weather, and NAM fields temporally aligned by timestamp. If a record is missing, we retain a mask and pass a neutral input so the fusion module can down-weight that modality. Satellite data are cropped as fixed 10×10 patches centered on the site, NAM features are sampled from four nearby grid nodes. We compute deterministic clear-sky irradiance to keep daylight samples only and require each training time to have a minimum fraction of valid future horizons (coverage filter) [21].

Inputs are normalized as follows: images/satellite use per-channel ImageNet mean/std; irradiance, surface weather, and NAM features are z-scored with statistics fit on the training split and applied to validation/test sets. Per-horizon target standardization is used for training and inverted for reporting. We append sinusoidal time encodings (time-of-day, day-of-year) to the ground time series, exclude any unavailable future targets from loss/metrics via masks, and precompute/cache vision embeddings for sky and satellite to decouple training from heavy backbones. After filtering, we use a chronological 70/15/15 train/val/test split and predict 12 horizons (1–24 h at 5-min base step), with condition-stratified evaluation (clear/partial/overcast) reported for analysis.

Non-stationarity mitigation. We mitigate diurnal and seasonal non-stationarity using three complementary mechanisms: (i) explicit cyclical time encodings (sin/cos of time-of-day and day-of-year) appended to the ground time-series inputs, (ii) a clear-sky-aware residual target in clearness-index space (§2.2), which forecasts deviations from current atmospheric transmissivity rather than raw irradiance magnitude, and (iii) adaptive fusion and meteorology-driven dynamic graph construction (§2.3) that re-weights modality contributions and spatial dependencies as weather regimes evolve. Together, these design choices reduce sensitivity to deterministic solar-geometry-driven magnitude variations and stabilize learning across seasons and lead times.

2.2 Feature Encoding and Clear-Sky Residual Representation

We encode each modality with a backbone matched to its structure and scale, producing fixed-width embeddings at the current decision time t from a look-back window of length L . Vision encoders operate frame wise at each timestamp, while sequence encoders summarize the past window into a single representation. The outputs are passed to the fusion module (presented in §2.3); the target is expressed in a clear-sky-aware residual space to reduce non-stationarity. The supervised target is defined in a clear-sky-aware residual space to reduce diurnal and seasonal non-stationarity, and we map predictions back to absolute irradiance for reporting. Below, we explain the details of the vision and sequence backbones in detail:

Vision backbone: We employed two vision encoders f_{θ}^{sky} and f_{θ}^{sat} for sky images and satellite patches. The former has allowed us to convert the normalized fisheye frame at time t into a compact embedding as (7):

$$h_t^{sky} = f_{\theta}^{sky}(X_t^{sky}) \in R^{d_v}, \quad d_v = 768 \quad (7)$$

where X_t^{sky} is the sky image at time t , θ denotes learnable parameters, and d_v is the vision embedding dimension. This encoder captures fine-grained cloud texture and local motion cues salient for short-term ramps. The satellite encoder processes the temporally aligned site-centered patch:

$$h_t^{sat} = f_{\theta}^{sat}(X_t^{sat}) \in R^{d_v}, \quad d_v = 768 \quad (8)$$

where X_t^{sat} is the satellite patch at time t . These embeddings summarize mesoscale cloud fields and upstream advection beyond the sky camera’s field of view.

Sequence backbone: We employed a temporal convolutional network (TCN) [22] with dilated 1-D convolutions to encode the past window of surface measurements (irradiance and weather) augmented with cyclical time features. Let $\tilde{X}_{t-L+1:t}^{ts} \in R^{L \times F}$ be the standardized sequence over the look-back window, where F is the number of ground time-series features and $(\cdot)$ indicates z-scoring using training-split statistics. The TCN output is global-average-pooled over time:

$$h_t^{ts} = GAP(TCN_{\theta}(\tilde{X}_{t-L+1:t}^{ts})) \in R^{d_t}, \quad d_t = 192 \quad (9)$$

where $GAP(\cdot)$ denotes global average pooling across the temporal dimension, and d_t is the time-series embedding dimension. Equation (9) summarizes sequential fluctuations and slower trends in the grid site's atmospheric state.

We retain the standardized NAM tensor $\tilde{X}_{t-L+1:t}^{NAM} \in R^{L \times G \times F_n}$ (window $\times$ grid-nodes $\times$ variables) without collapsing it at this stage, where G is the number of NAM nodes and F_n is the number of NAM variables. Its spatio-temporal representation is learned by the dynamic graph module in §2.3. This preserves node-wise structure needed to compute time-varying, meteorology-driven adjacency. **Table 1** lists encoder specifications.

Table 1. Encoder Architecture Specifications

Component	Parameter	Value
ViT Encoders (Sky & Sat)	Backbone Model	vit_b_16 (ViT-Base)
	Input Resolution	224×224 pixels
	Patch Size	16×16 pixels
	Output Embedding	768 dimensions
TCN Encoder (Time Series)	Kernel Size	5
	Number of Blocks	4
	Dilation Factors	[1, 2, 4, 8]
	Output Embedding	192 dimensions

Clear-sky residual target and optimization. Let y_t denote observed GHI at time t and c_t the deterministic clear-sky GHI computed for the same timestamp (used to filter daylight samples). We define the clipped clearness index as follows (10)

$$k_t = clip\left(\frac{y_t}{c_t}, 0, k_{max}\right), k_{max} = 1.2, \quad (10)$$

where $clip(x, a, b)$ clamps x to $[a, b]$, and k_{max} bounds physically plausible *cloud enhancement* effects and improves numerical stability. For a prediction horizon $h \in \{1, \dots, H\}$, the supervised target is the residual in clearness-index space:

$$r_{t,h} = clip(k_{t+h} - k_t, -k_{max}, k_{max}) \quad (11)$$

where k_{t+h} is the future clearness index at timestamp $t+h$, and H is the number of forecast lead times.

During training, we standardize targets per horizon using training-split statistics: $\tilde{r}_{t,h} = (r_{t,h} - \mu_h)/\sigma_h$, where μ_h and σ_h are the mean and standard deviation of $r_{t,h}$ for each horizon h computed on the training set. The model predicts $\hat{\tilde{r}}_{t,h}$, and we optimize a masked robust regression loss (Smooth - L_1) over available daylight horizons:

$$\mathcal{L}_{pred} = \frac{1}{\sum_{t,h} m_{t,h}} \sum_{t,h} m_{t,h} SmoothL1(\hat{\tilde{r}}_{t,h} - \tilde{r}_{t,h}), \quad (12)$$

where $m_{t,h} \in \{0,1\}$ masks non-daylight timestamps and any missing future targets, and $SmoothL1(\cdot)$ is the robust loss.

At inference, we invert standardization and map residual predictions back to absolute GHI via

$$\hat{k}_{t+h} = clip(\hat{r}_{t,h} + k_t, 0, k_{max}), \hat{y}_{t+h} = \hat{k}_{t+h} \cdot c_{t+h}. \quad (13)$$

where $\hat{r}_{t,h}$ is the de-standardized residual prediction and c_{t+h} is the clear-sky GHI at the future timestamp. This residual formulation reduces diurnal/seasonal non-stationarity by forecasting deviations from the current atmospheric transmissivity rather than raw irradiance magnitude.

2.3 Adaptive Multimodal Fusion with Dynamic GNN

We combine modality-specific embeddings with a spatio-temporal graph representation that adapts to evolving meteorology. The goal is to (i) propagate context across nearby sites using a dynamic graph built from mesoscale fields and (ii) fuse image, satellite, time-series, and graph features with per-sample weights that reflect their instantaneous relevance [23].

We have formulated a dynamic spatio-temporal graph over NAM features as follows. Let $H_\tau^{NAM} \in R^{G \times F_n}$ denote standardized NAM features for G nodes at time $\tau \in \{t - L + 1, \dots, t\}$. We build a time-varying sequence graph $G_\tau = (V, E_\tau)$ with adjacency A_τ computed from cosine similarity of L_2 – normalized node features. A spatial graph attention layer updates node embeddings using A_τ as a soft connectivity prior:

$$U_\tau^{(l+1)}(i) = \sigma \left(\sum_{j \in V} \alpha_{ij,\tau}^{(l)} W^{(l)} U_\tau^{(l)}(j) \right), \alpha_{ij,\tau}^{(l)} = softmax_j \left(\varphi \left(U_\tau^{(l)}(i), \left(U_\tau^{(l)}(j) + \beta A_\tau(i,j) \right) \right) \right) \quad (14)$$

where $U_\tau^{(l)}(i)$ is the embedding node i at GNN layer l and time τ , $W^{(l)}$ is a learnable linear projection, $\sigma(\cdot)$ is a nonlinearity, $\alpha_{ij,\tau}^{(l)}$ is the attention coefficient from node j to node i , and $\varphi(\cdot)$ is learnable attention scoring function. $A_\tau(i,j)$ denotes the dynamic adjacency weight between nodes i and j at time τ , and β scales the meteorology-driven bias inside attention. We summarize the spatio-temporal graph outputs with pooling over nodes and time to obtain a compact graph representation:

$$h_t^{st} = Pool_{\tau,i}(GNN(H_\tau^{NAM}, A_\tau)) \in R^{d_s} \quad (15)$$

where $Pool_{\tau,i}(\cdot)$ denotes pooling across time τ and nodes i , and d_s is the graph-token embedding dimension.

Then from §2.2, we have $h_t^{sky} \in R^{d_v}$, $h_t^{sat} \in R^{d_v}$, $h_t^{ts} \in R^{d_t}$, and $h_t^{st} \in R^{d_s}$. We project each to a shared d -dimensional space using following (16):

$$z_t^m = P_m h_t^m \in R^d, m \in \{sky, sat, ts, st\} \quad (16)$$

where $P_m \in R^{d \times d_m}$ is a learnable projection matrix for modality m ($d_m \in \{d_v, d_t, d_s\}$).

Next, we compute a per-sample gating vector $\alpha_t \in R^4$ that assigns an importance weight to each modality at time t . A lightweight gating network $g(\cdot)$ takes the concatenated projected tokens and outputs gate logits, which are converted to non-negative weights via softmax:

$$\alpha_t = softmax(g([z_t^{sky}; z_t^{sat}; z_t^{ts}; z_t^{st}])). \quad (17)$$

where $\alpha_{t,m} \geq 0$ and $\sum_m \alpha_{t,m} = 1$. If a modality is unavailable at time t , we suppress its logit using an availability mask, forcing its weight to zero. The fused representation is:

$$z_t^{fuse} = \sum_m \alpha_{t,m} z_t^m. \quad (18)$$

This adaptive fusion enables the model to prioritize the most informative modalities under the prevailing conditions. The fused token z_t^{fuse} is then passed to the horizon-aware MoE head (§2.4) to generate multi-horizon forecasts.

2.4 Horizon-Aware MoE

The fused token z_t^{fuse} is mapped to multi-horizon forecasts using a mixture-of-experts head composed of E lightweight MLP experts. Each expert $e \in \{1, \dots, E\}$ produces a horizon vector $\hat{r}_t^{(e)} \in R^H$ in the same residual space defined in §2.2. A gating network computes non-negative mixture weights $\pi_t \in R^E$ (summing to one) from the fused representation:

$$\pi_t = \text{softmax}\left(\frac{W_g z_t^{fuse}}{\tau_g}\right) \quad (19)$$

where W_g are learnable gate parameters and τ_g is a temperature hyperparameter controlling routing entropy. The final MoE output is the convex combination:

$$\hat{r}_t = \sum_{e=1}^E \pi_{t,e} \hat{r}_t^{(e)} \quad (20)$$

Training strategy (Gumbel-Softmax routing). During training, we employ a Gumbel–Softmax relaxation to encourage sharper expert specialization while maintaining end-to-end differentiability [24]. At inference time, routing reduces to the standard softmax in (19), and the MoE head produces all horizons jointly in a single forward pass.

Load balancing to prevent expert collapse. A common issue in MoE training is expert collapse, where only a small subset of experts receives most samples. To promote balanced utilization and stable specialization, we add a batch-level load-balancing regularizer that encourages uniform average gate usage across a mini-batch of size B . Let $\pi_{b,e}$ denote the gate weight assigned to expert e for sample b . We define the batch-average expert usage:

$$\hat{\pi}_e = \frac{1}{B} \sum_{b=1}^B \pi_{b,e} \quad (21)$$

and the balancing loss:

$$L_{bal} = \sum_{e=1}^E \left(\hat{\pi}_e - \frac{1}{E}\right)^2 \quad (22)$$

The overall training objective combines the masked regression loss L_{pred} with the balancing term:

$$L = L_{pred} + \lambda L_{bal} \quad (23)$$

where λ controls the strength of the balancing regularization. This term keeps gradients active for under-utilized experts, reduces collapse, and improves routing stability and expert specialization without increasing inference cost.

3. Experimental Setup

3.1 Dataset

To validate our proposed approach, we use the publicly available multi-year Folsom dataset [25], which integrates four synchronized modalities for solar forecasting development and benchmarking, including hemispheric fisheye sky images from 2014–2016 that capture local

cloud dynamics, ground-based time series of irradiance and surface meteorology, 10×10 satellite tiles centered on the site that summarize mesoscale cloud motion, and NAM mesoscale model fields extracted from four nearby grid nodes, encompassing wind components, temperature, humidity, pressure, cloud cover, precipitation, and downwelling shortwave flux. Forecasts span 1–120 hours ahead, with 5-minute resolution up to 24 hours (12–288 steps) and hourly thereafter, while summary results report horizons at $\{1, 2, 4, 8, 12, 24, 36, 48, 60, 72, 96, 120\}$ hours, with daylight masking applied.

Fig. 2 shows how sky conditions affect GHI and image features: Clear skies yield high GHI, low mean RGB/luminance (uniform blue), and low std (low contrast). Cloudy skies reduce GHI, increase mean values (brighter whites), and raise std (textured variability). This highlights how cloud dynamics drive fluctuations in solar irradiance and image statistics, essential for forecasting models.

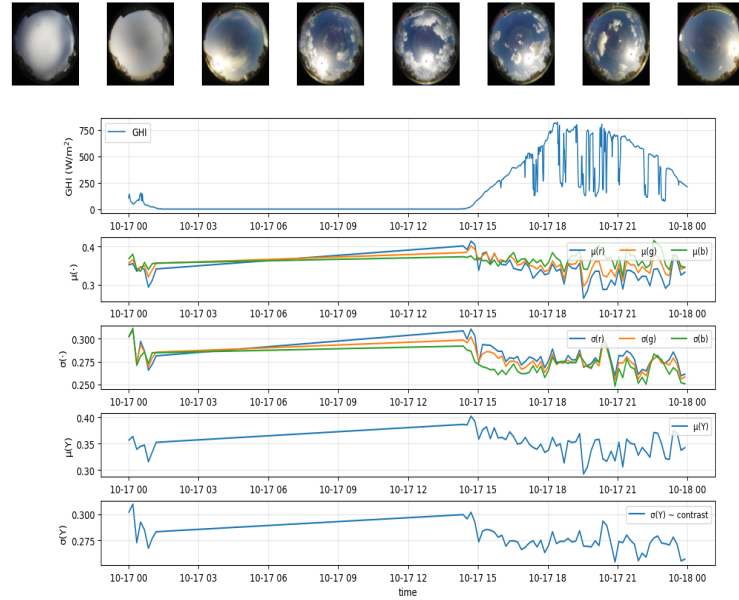


Fig. 2. Example visualization from the Folsom dataset, demonstrating the impact of sky conditions on irradiance and image-derived features.

3.2 Evaluation metrics and baselines

We evaluate accuracy per horizon h in absolute irradiance units (W m^{-2}). Let y_{t+h} and $\hat{y}_{t+h}$ be the true and predicted GHI for decision time t , and let $M_{t,h} \in \{0,1\}$ mask out non-daylight or missing targets. The masked mean-squared error (MSE) and root-mean-squared error (RMSE) are represented with (24):

$$MSE_h = \frac{\sum_t M_{t,h} (y_{t+h} - \hat{y}_{t+h})^2}{\sum_t M_{t,h}}, \quad RMSE_h = \sqrt{MSE_h} \quad (24)$$

We also report per-horizon mean absolute error (MAE) $MAE_h = \frac{\sum_t M_{t,h} |y_{t+h} - \hat{y}_{t+h}|}{\sum_t M_{t,h}}$, and horizon-pooled “global” RMSE by aggregating all valid (t,h) pairs into one set before computing RMSE.

To contextualize error, we compute a skill score against a persistence baseline evaluated on the same masked sample set:

$$Skill = \left(1 - \frac{RMSE_{model}}{RMSE_{pers}}\right) \times 100\% \quad (25)$$

To evaluate performance, we benchmark our model against baseline methods and two persistence forecasts: simple persistence of Global Horizontal Irradiance (GHI) and smart persistence of the clearness index (k_t). Forecasting skill relative to these benchmarks is aggregated across intra-hour, intra-day, and day-ahead horizons. To assess robustness, performance metrics are stratified by sky conditions (clear, partially cloudy, overcast) and during high-variability ramp events.

To ensure a fair and robust comparison, all baseline models were trained using the identical data splits (70/15/15 chronological split), preprocessing, and masked Smooth-L1 loss function as our proposed model. For the classic linear models, we used *LassoCV* and *RidgeCV*, which automatically perform hyperparameter tuning via 10-fold cross-validation to find the optimal regularization strength. For deep learning baselines (LSTM, TCN, Informer, Crossformer, PatchTST), we adopted recommended settings from the original publications and widely-used open-source implementations, with minor adjustments for fair comparison across models. The specific architectural hyperparameters for each baseline are detailed in [Table 2](#).

Table 2. Baseline Model Hyperparameters

Model	Hyperparameter	Value	Model	Hyperparameter	Value
Lasso	α	10-fold CV	Informer	d_model	128
Ridge	α	10-fold CV		n_heads	4
LSTM	hidden_dim	32		encoder_layers	2
	num_layers	2		ff_dim	256
	optimizer	Adam	Crossformer	d_model	128
TCN	num_filters	64		n_heads	4
	kernel_size	3		encoder_layers	2
	dilation_factors	[1,2,4,8]		ff_dim	256
	dropout_rate	0.2			
PatchTST	d_model	128			
	n_heads	4			
	encoder_layers	2			
	patch_len	3			
	ff_dim	256			

Informer, Crossformer, and PatchTST hyperparameters were adapted from official implementations and scaled to ensure computational fairness and stable training across baselines. Higher-capacity configurations reported in original papers (e.g., Informer d_model=512, 8 heads, deeper layers) were computationally heavy for our scenario and showed marginal performance gains relative to increased cost.

The analysis is completed by an ablation study (Section 5.1) and a comparative performance review against state-of-the-art (SOTA) models (Section 5.2).

3.3 Training

We have trained with *AdamW* using a learning rate 3×10^{-4} and weight decay of 10^{-4} . Dropout is 0.2 on the fused token. Gradients are clipped at norm 1.0. Mixed precision is enabled. We apply dropout of 0.2 on the fused token and clip gradients at an ℓ_2 norm of 1.0. The learning rate follows a *ReduceLROnPlateau* schedule that halves the rate after five validation epochs without improvement. Early stopping monitors validation loss with a patience of twelve epochs. The batch size is 128. To reduce overhead, vision features for sky and satellite are

precomputed and cached so the main training loop optimizes the fusion, graph, and output heads. We fix a random seed for data splitting, initialization, and data loading to ensure reproducibility. After daylight and coverage filtering, we use a chronological split of 70% for training, 15% for validation, and 15% for testing. We selected the checkpoint with the lowest validation loss for all reported test results.

3.4 Hyperparameter tuning

We performed a *coarse-to-fine grid search* on the validation split, holding data splits and masks fixed. Each setting was trained with early stopping, and the selection criterion was the average validation RMSE across all horizons, with ties broken by global RMSE and latency. Final results are reported from the checkpoint with the best validation score. The grid search hyperparameter ranges are summarized in [Table 3](#).

Table 3. Summary of hyperparameters

Hyperparameter	Values	Hyperparameter	Values
Input window L	[48, 96, 144, 192]	MoE experts E	[3, 4, 5, 7]
Dropout p (fused token)	[0.0, 0.1, 0.2, 0.3]	Gate temperature τ	[0.5, 0.8, 1.0, 1.5]
Learning rate	[1e-3, 5e-4, 3e-4, 1e-4]	Load-balancing λ	[1e-3, 5e-3, 1e-2, 5e-2]
Weight decay	[0, 5e-5, 1e-4, 5e-4]	Satellite crop tiles	[8, 10, 12, 14]
GNN layers	[1, 2, 3, 4]	Fusion embedding d	[256, 384, 512, 768]
Dynamic-graph bias β	[0.0, 0.5, 1.0, 2.0]		

[Fig. 3](#) illustrates the accuracy–efficiency trade-off as the look-back window length L is increased. Validation accuracy improves steadily as L grows from short histories to a 12-hour context, and the best mean validation RMSE is achieved at $L=144$ (12 hours). Extending the look-back beyond this point does not yield further gains; instead, performance shows a mild deterioration for longer windows (e.g., $L=192$ and $L=288$). In contrast, computational cost increases monotonically with L : longer histories require more training time and incur higher inference latency. Taken together, [Fig. 3\(a\)](#) provides direct evidence that $L=144$ offers the most favorable accuracy–efficiency balance, and we therefore adopt it as the default input window in the final configuration.

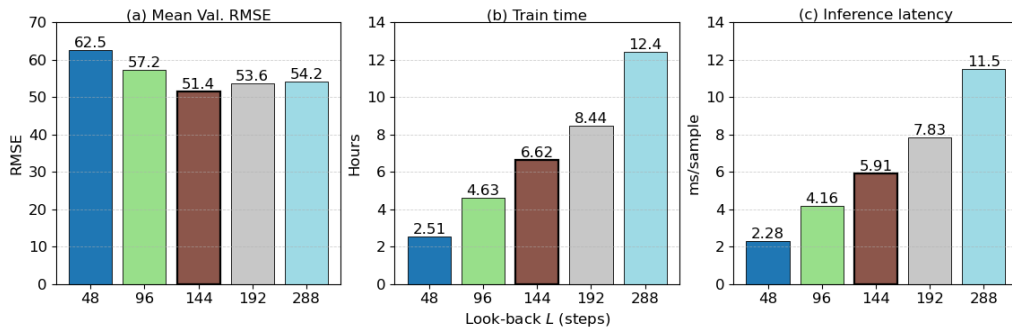


Fig. 3. Effect of look-back window length L on (a) mean validation RMSE, (b) training time, and (c) single GPU inference latency

Beyond the look-back length, hyperparameter tuning favored a balanced capacity–latency setting. A fusion embedding of $d=512$ provided strong validation performance without the runtime overhead of larger widths, while $p=0.2$ dropout yielded stable generalization. We obtained the most reliable optimization behavior with $LR=3 \times 10^{-4}$ and $WD=10^{-4}$. For the graph module, a two-layer GNN and a moderate meteorology bias ($\beta=1.0$) improved robustness

without introducing noisy connectivity. In the MoE head, $E=5$ with $\tau=1.0$ performed consistently across horizons, and $\lambda=10^{-2}$ mitigated expert collapse. A 10×10 satellite crop preserved adequate mesoscale context at low cost. For the final configuration ($L=144$), the end-to-end model forward-pass latency is 5.91 ms per sample on a single GPU, including the dynamic graph update, adaptive fusion, and MoE prediction. Relative to the 5-minute operational range typically used in scheduling pipelines, this runtime is comfortably within real-time constraints. Fig. 3(c) reports how inference latency scales with L across all evaluated window lengths.

4. Experimental results and analysis

4.1 Training Dynamics and Convergence

We optimize the model end-to-end with AdamW and a masked Smooth-L1 objective computed on standardized clearness-index residual targets (daylight only). The loss is averaged over all valid horizons in each batch; a binary mask excludes night and missing horizons. Regularization includes dropout on the fused token, feature dropout on inputs, load-balancing for the MoE, and gradient-norm clipping (1.0). To stabilize and accelerate training we use mixed precision and a ReduceLROnPlateau scheduler that halves the learning rate when the validation loss stalls.

Fig. 4 shows the training and validation losses over epochs (5-epoch moving average). The curves exhibit three characteristic phases. (i) *Fast descent* over the first ~ 20 – 30 epochs as the model fits coarse structure in the residual target. (ii) *A slower improvement regime* from ~ 30 to ~ 60 epochs as the network refines spatial-temporal couplings and the fusion gate stabilizes. (iii) *Scheduler-driven fine-tuning* after learning-rate reductions at epochs 60, 100, and 110: each LR cut is followed by a visible drop in both train and validation curves, indicating effective exploitation of additional capacity at smaller step sizes. Throughout training the generalization gap remains small (validation slightly above training), consistent with the use of residual targets, adaptive fusion, and strong regularization.

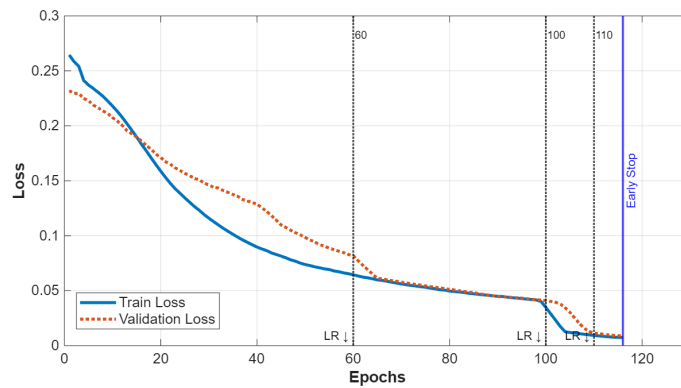


Fig. 4. Training and validation loss versus epoch, averaged with a 5-epoch moving window. Vertical black dotted lines mark learning-rate reductions triggered by ReduceLROnPlateau at epochs 60, 100, and 110 (annotated “LR ↓”). The blue vertical line labeled Early Stop denotes the early-stopped checkpoint at epoch 116, which is used for all test-set results.

4.2 Per-Horizon Error Metrics

We evaluate forecast accuracy at different horizons using the error metrics defined in Eqs. (19–20): RMSE, MAE, and skill score against persistence (persKT) and clear-sky baselines. To ensure robustness, we report two complementary evaluation styles: aggregated (pooled over all valid time–horizon pairs) and averaged (computed per-timestep and then averaged). Aggregated metrics emphasize magnitude-sensitive errors, while averaged metrics highlight relative performance across horizons. Both perspectives are complementary and provide a more complete assessment of forecasting quality.

4.2.1 Aggregated Per-Horizon Metrics

The aggregated, per-horizon results provide a comprehensive view of the model's forecasting accuracy (Table 4) for Global Horizontal Irradiance (GHI), with the metrics emphasizing performance during high-intensity midday periods crucial for solar energy applications. To quantify performance, $RMSE_{pred}$ is used to heavily penalize large errors, which are the most impactful, while MAE_{pred} offers a more direct interpretation of the average error magnitude. Beyond these absolute figures, $skill$ scores are essential for benchmarking the model's value. The results show a strong, positive skill score against a persistence baseline $Skill_{pred}$ across all horizons, peaking at 46.5%, which confirms the model's robust ability to capture weather dynamics better than a simple "no-change" assumption. In contrast, the skill score against a clear-sky baseline ($Skill_{clear}$) reveals a more nuanced story. While the primary goal is positive skill, analyzing the negative results is crucial as it defines the model's specific weaknesses. The model initially underperforms this idealized benchmark at very short horizons because predicting the exact timing of intermittent clouds is extremely difficult. However, this initial underperformance serves to highlight the model's true strengths, as the skill score transitions to become strongly positive (up to 54.9%) at mid-to-long horizons (12–120 hours).

Table 4. Aggregated per-horizon error metrics for GHI forecasting (prediction vs persistence and clear-sky baselines).

Horizon (hr)	N	MSE_{pred} (W/m^2) ²	$RMSE_{pred}$ (W/m^2)	MAE_{pred} (W/m^2)	$Skill_{pred}$ (%)	$RMSE_{persKT}$ (W/m^2)	$RMSE_{clear}$ (W/m^2)	$Skill_{clear}$ (%)
1	100,519	3,727	61.1	42.3	16.8	73.3	123.7	-68.6
2	93,935	5,111	71.5	48.9	19.0	88.3	124.8	-41.4
4	84,622	6,076	77.9	52.3	22.5	94.6	123.7	-23.0
8	74,451	6,419	80.1	53.8	26.7	109.4	117.9	-7.9
12	64,251	6,345	79.7	53.0	29.5	113.1	106.4	5.8
24	54,051	5,104	71.4	48.0	34.1	108.3	89.1	17.9
36	43,851	4,030	63.5	43.1	37.7	101.7	70.1	31.1
48	33,651	3,054	55.3	37.6	40.8	93.3	53.5	42.7
60	15,497	1,364	36.9	26.1	46.5	68.9	31.1	54.9
72	9,147	431	20.8	15.4	29.0	29.3	19.1	34.7
96	26,021	2,620	51.2	36.0	36.5	80.6	47.9	40.6
120	101,936	9,881	99.4	70.1	18.8	122.4	121.8	0.5

4.2.2 Averaged Per-Horizon Metrics

In this evaluation step, the model using per-horizon metrics that are averaged over time. This method is specifically designed to provide a more balanced assessment of performance across all daylight conditions. Unlike the aggregated approach, it gives equal weight to predictions made during lower-irradiance periods (such as mornings, evenings, or overcast days) and high-intensity midday sun, offering a clearer picture of the model's overall fidelity.

Table 5 presents the averaged per-horizon error metrics for GHI forecasting, highlighting the differences between our model's predictions and the two baselines (persistence and clear-sky). The results show that the proposed model maintains consistently positive skill scores relative to persistence across nearly all horizons, confirming its ability to capture short-term variability and outperform a simple no-change forecast. More importantly, when compared against the clear-sky baseline, the averaged evaluation reveals a clear advantage of our approach: at shorter horizons (1–12 h), the model achieves strongly positive skill values (up to 47.8% at 1 h), demonstrating its capacity to accurately account for cloud cover and variable weather, where the clear-sky assumption fails. This advantage is especially critical under low-irradiance conditions (mornings, evenings, or cloudy periods), which are appropriately weighed in this averaged analysis. While skill against clear-sky diminishes and becomes negative beyond 48 h due to the increasing difficulty of predicting cloud cover at longer horizons, the model still provides meaningful improvements over persistence. These results underline the strengths of the proposed approach: robust short- to mid-term forecasting under variable atmospheric conditions, and a more balanced performance profile than either persistence or clear-sky baselines.

Table 5. Average per-horizon error metrics for GHI forecasting (prediction vs persistence and clear-sky baselines).

Horizon (hr)	N	MSE_{pred} (W/m^2) ²	$RMSE_{pred}$ (W/m^2)	MAE_{pred} (W/m^2)	$Skill_{pred}$ (%)	$RMSE_{persKT}$ (W/m^2)	$RMSE_{clear}$ (W/m^2)	$Skill_{clear}$ (%)
1	171	4083.9	45.0	31.3	18.9	55.5	86.2	47.8
2	171	5544.2	53.4	38.5	23.9	70.2	85.5	37.6
4	171	6632.3	60.5	44.1	27.3	83.2	84.7	28.5
8	170	7072.7	64.4	47.1	30.5	92.6	82.1	21.6
12	170	6888.5	66.1	48.0	31.1	95.9	75.0	11.9
24	170	5418.1	59.6	42.8	33.9	90.2	64.8	7.9
36	170	4040.2	52.1	36.9	35.5	80.7	52.4	0.6
48	170	2641.9	40.5	27.8	38.9	66.2	37.3	-8.3
60	114	987.1	24.0	16.4	44.5	43.3	18.4	-30.5
72	65	305.2	12.3	7.8	24.7	16.3	12.0	-2.8
96	146	2062.4	32.6	22.1	36.8	51.5	29.2	-11.6
120	168	11360.6	81.5	67.8	12.5	93.1	85.9	5.1

4.3 Multi-Horizon Forecast Evaluation

To further assess the quality of our forecasting model, we examine prediction performance across a range of horizons from 1 hour to 120 hours ahead. While error metrics such as RMSE and MAE (Section 4.2) provide quantitative summaries, they do not fully capture the temporal structure of the forecasts. Therefore, we visualize representative trajectories comparing ground truth and model predictions over contiguous time periods.

Fig. 5 presents actual vs. predicted global horizontal irradiance (GHI) for selected horizons: {1, 2, 4, 8, 12, 24, 36, 48, 60, 72, 96, 120 hours}. Each subplot corresponds to a fixed forecast horizon, and the forecasts are evaluated over approximately 300 consecutive timesteps. This arrangement highlights how the model performs at both short-term and long-term scales. At shorter horizons (1–4 h), the model closely tracks rapid fluctuations in GHI, including cloud-driven variability, with minimal lag. Mid-range horizons (8–24 h) exhibit stronger smoothing effects, where the diurnal shape is preserved but finer variability is underestimated. For longer horizons (36–120 h), the model remains stable and continues to capture the solar cycle trend, though amplitude mismatches become more pronounced due to accumulated uncertainty.

Overall, these results demonstrate that the proposed model is capable of producing consistent forecasts across a wide temporal span. In particular, the ability to retain diurnal dynamics up to 120 hours ahead suggests robustness for practical applications such as energy scheduling and grid integration, while short-term predictions remain competitive for intra-day operational forecasting. To evaluate the temporal performance of our forecasting model, we visualize actual vs. predicted GHI trajectories across multiple horizons ranging from 1 hour to 120 hours ahead.

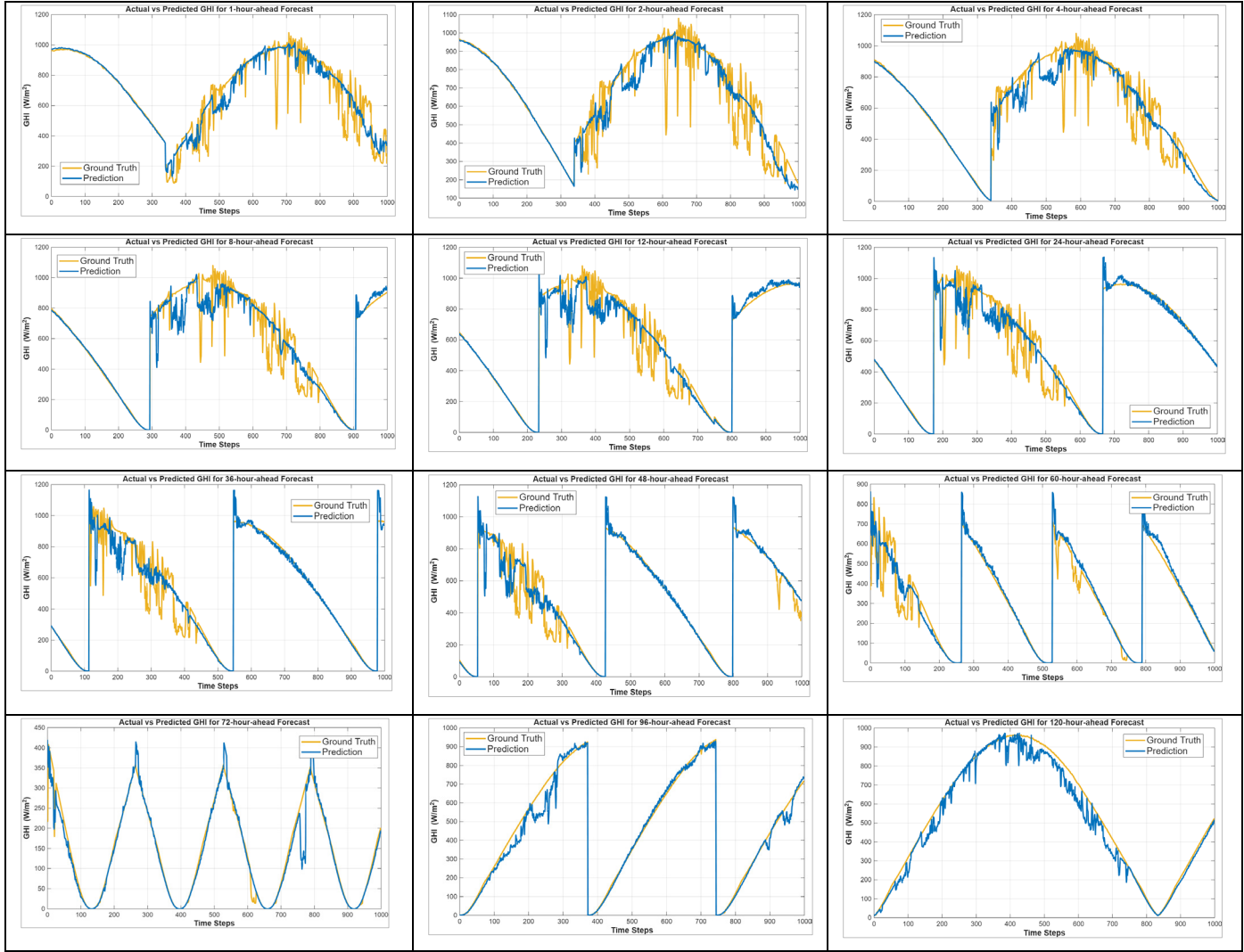


Fig. 5. Actual vs. predicted Global Horizontal Irradiance (GHI) across horizons from 1 to 120 hours in a 3×4 matrix. Yellow denotes ground truth, blue shows model forecasts. Short horizons (1–12 h) track variability well, while longer horizons (24–120 h) show gradual error growth but preserve diurnal cycles.

4.4 Sky-Conditioned Evaluation

Traditional aggregated metrics mix together diverse sky conditions, masking the differences in model behavior under stable versus highly variable irradiance regimes. To provide a more nuanced evaluation, we stratify errors by *Open-Sky*, *Partial Cloudy*, *Overcast*, and *Rampy*

conditions. **Fig. 6** illustrates the per-timestamp prediction errors for each condition at the 1-hour horizon, comparing the forecasting model (*pred error*) against *persistence* (*pers error*) and *clear-sky* (*clear error*) baselines. The analysis of per-timestamp errors reveals how the proposed model's predictive error (blue) compares against a naïve smart persistence baseline (orange) and a clear-sky reference (green) across various weather conditions. Under open-sky conditions, all error types remain low, confirming the model's stability. In partially cloudy skies, the model consistently reduces error compared to the persistence baseline, demonstrating its ability to handle moderate variability. During overcast periods, the clear-sky model produces massive errors by overestimating irradiance, whereas the proposed model aligns well with the low-light ground truth. The most challenging rampy conditions cause large error oscillations for the persistence baseline, which fails to adapt to rapid changes. In contrast, the proposed forecasting model, while still challenged, successfully mitigates these errors, proving its superior performance in adapting to volatile cloud dynamics.

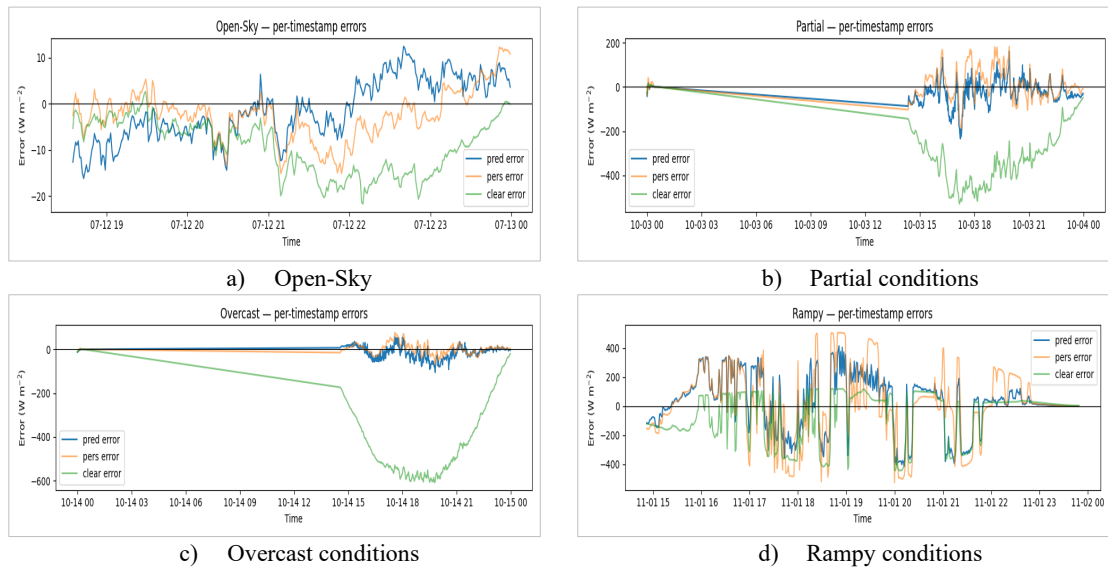


Fig. 6. Per-timestamp forecast errors for a 1-hour horizon, stratified by four distinct sky conditions.

Fig. 7 compares the RMSE of our proposed fusion-based forecasting model (“Prediction”) against persistence (KT) and clear-sky baselines across four distinct environmental conditions: open-sky, partial, overcast, and rampy. The results demonstrate that our fusion algorithm consistently adapts to diverse irradiance regimes and maintains competitive performance under both stable and highly variable conditions. In simple open-sky and stable overcast scenarios, our model matches the strong performance of persistence while easily outperforming the clear-sky model, confirming its reliability in predictable conditions.

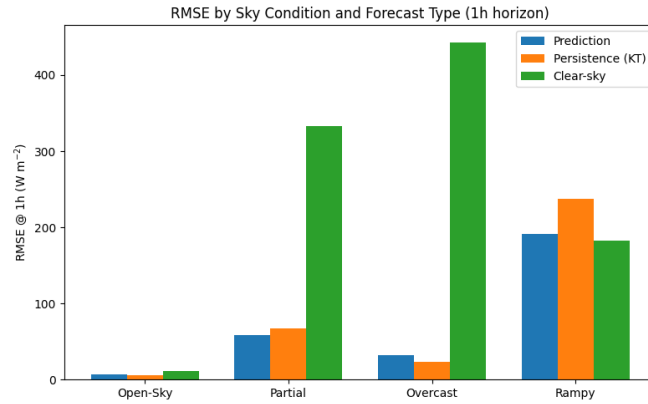


Fig. 7. RMSE comparison under different sky conditions (1h horizon). The proposed fusion-based prediction model demonstrates consistent robustness, outperforming persistence and clear-sky baselines particularly in partial and rampy regimes where irradiance variability is highest.

The model’s true strength is revealed in more challenging regimes. Under partially cloudy skies, it significantly reduces errors by effectively learning and adapting to cloud-induced variability. Most critically, during highly volatile rampy conditions, our fusion algorithm substantially outperforms both baselines by integrating diverse data sources to better anticipate sudden irradiance changes. This evaluation confirms that our model is not only accurate in aggregate but also consistently adapts to a wide spectrum of real-world conditions, making it a more reliable solution for operational solar forecasting.

5. Ablation Study and Comparative Analysis

5.1 Ablation Study

To rigorously evaluate the contribution of each architectural component, we conducted a controlled ablation study. The study was designed to isolate the impact of each sub-module by systematically removing or replacing it while holding all other experimental conditions constant, including the optimizer, learning rate schedule, data splits, loss function, and forecast horizons. We evaluated the following six ablated variants against our full proposed model:

- 1) **No NWP:** The complete North American Mesoscale (NAM) weather prediction feature set is removed from the model's input.
- 2) **No STGNN:** The Spatio-Temporal Graph Neural Network (ST-GNN) is replaced with a simple mean-pooling operation over the NAM features to assess the value of structured modeling.
- 3) **No Vision:** Both sky and satellite image-based features are removed, forcing the model to rely solely on numerical data.
- 4) **No Sky Images:** Only the ground-based sky camera features are removed.
- 5) **No Satellite:** Only the satellite imagery features are removed.
- 6) **No MoE Head:** The Mixture-of-Experts (MoE) prediction head is replaced with a standard Multi Layer Perceptron (MLP) of comparable parameter count.

Each variant was trained under identical conditions, and their validation loss curves are presented in **Fig. 8**. The analysis reveals a clear performance hierarchy and distinct learning dynamics, underscoring the synergistic effect of the full model. Our full model (*Ours*) achieves the lowest validation loss at approximately 0.065. Removing the *MoE* head results

in the smallest degradation (loss ≈ 0.095), followed by the removal of sky images (≈ 0.10) and satellite imagery (≈ 0.14). More significant performance losses occur when removing all vision (≈ 0.19) or replacing the STGNN with mean pooling (≈ 0.20). The most substantial degradation is observed in the No NWP variant, whose final loss of ≈ 0.33 is approximately five times higher than the full model.

The learning dynamics further illuminates each component's role. The performance gap is evident early in training (≈ 20 epochs), where our full model already achieves a loss of ≈ 0.14 , far outpacing the structurally critical variants like No STGNN (≈ 0.30) and No NWP (≈ 0.38). The training curves also reveal distinct behaviors: the No Vision model plateaus early, lacking the fine-grained cues for later optimization; the No STGNN model exhibits a mid-epoch stall, showing an inability to propagate forecast information effectively; and the No NWP model improves at a persistently slow rate, underscoring that large-scale meteorological context is essential for efficient learning.

To summarize, our proposed model demonstrates superior predictive accuracy by synergistically fusing long-range numerical weather predictions with high-frequency visual data through a specialized deep learning architecture. This multi-modal approach results in both faster convergence and a lower generalization error, as each component provides a distinct and indispensable contribution to the final, highly accurate forecast.

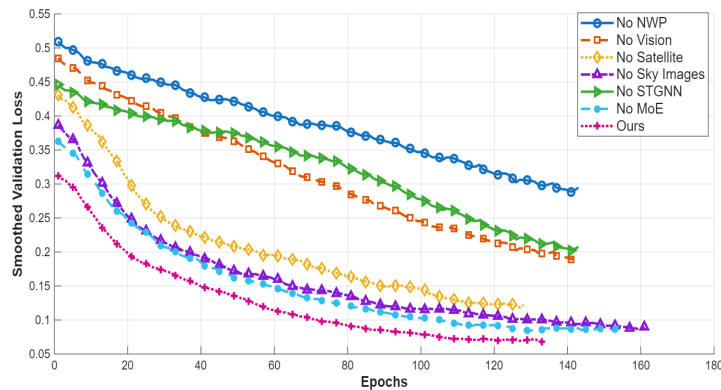


Fig. 8. Validation loss performance trends in the Ablation Study.

5.2 Comparative Analysis

To evaluate solar forecasting performance across different time scales, we implemented and compared a suite of machine learning and deep learning algorithms. The models are broadly categorized into two groups: Classic Linear Models, including Ordinary Least-Squares (OLS) and Lasso regression [26], and Deep Learning Models, which encompass Long Short-Term Memory (LSTM) networks, Temporal Convolutional Networks (TCN), and several state-of-the-art Transformer-based architectures (Informer [27], Crossformer [28], and PatchTST [29]). Our proposed model, denoted as "Ours*", is benchmarked against this diverse set. As discussed in Section 1, comparable state-of-the-art multimodal models that use our specific combination of four data sources are not readily available for a direct benchmark.

By following baseline paper implementation [25], models were trained using two distinct feature sets: a purely endogenous set, utilizing only historical irradiance data, and a combined set that includes exogenous features. The source of exogenous data varied by scenario, including sky-image features for intra-hour forecasts, satellite imagery for intra-day forecasts, and Numerical Weather Prediction (NWP) data for day-ahead forecasts. Performance is

evaluated using Root Mean Square Error (RMSE) and Forecast Skill, with the skill baseline being a Persistence model for short-term forecasts and the NAM model for day-ahead forecasts.

The results for intra-hour forecasting, summarized in **Table 6**, demonstrate the superior performance of our proposed model ("Ours*"). For GHI forecasting, our model achieves a skill of 18.10%, a significant improvement over the next-best models, LSTM (13.53%) and Crossformer (13.44%). The performance gap is even more pronounced for the more challenging DNI forecast, where our model's skill of 10.04% nearly doubles that of the next-best performers, TCN and Informer (both 5.69%). While the classic linear models remain competitive, the deep learning architectures and particularly our proposed model show a clear advantage in capturing the complex, short-term dynamics informed by sky imagery.

Table 6. Comparative analysis of intra-hour forecasting performance across Ours and competing approaches

Forecasting Horizon	Models	GHI		DNI	
		RMSE (W/m ²)	Skill (%)	RMSE (W/m ²)	Skill (%)
Intra-hour	Baseline	84.64	NaN	139.81	NaN
	OLS	77.21	8.77	133.60	4.44
	Lasso	76.80	9.26	132.60	5.16
	LSTM	73.19	13.53	132.31	5.37
	TCN	73.88	12.71	131.86	5.69
	PatchTST	73.54	13.11	132.68	5.10
	Crossformer	73.26	13.44	132.29	5.38
	Informer	74.70	11.74	131.85	5.69
	Ours*	69.32	18.10	125.15	10.04

As the forecast horizon extends to the intra-day scale (**Table 7**), the advantage of sophisticated deep learning models becomes more evident. The classic linear models exhibit the lowest skill, suggesting their limitations in handling longer time dependencies. The Transformer-based models, particularly PatchTST, show strong performance with a GHI skill of 15.41%. However, our proposed model ("Ours*") establishes a commanding lead, achieving a GHI skill of 23.43% and a DNI skill of 21.25%. This represents a relative skill improvement of over 50% for GHI and 63% for DNI compared to the next-best model, highlighting its exceptional capability in leveraging satellite data for medium-term forecasting.

Table 7. Comparative analysis of intra-day forecasting performance across Ours and competing approaches

Forecasting Horizon	Models	GHI		DNI	
		RMSE (W/m ²)	Skill (%)	RMSE (W/m ²)	Skill (%)
Intra-day	Baseline	89.56	NaN	183.48	
	OLS	83.07	7.95	178.13	3.43
	Lasso	82.27	8.39	174.79	4.53
	LSTM	80.40	10.23	169.64	7.55
	Informer	77.68	13.27	166.57	9.21
	TCN	78.41	12.44	163.08	11.12
	Crossformer	77.22	13.77	160.87	12.32
	PatchTST	75.76	15.41	159.59	13.01
	Ours*	68.58	23.43	144.49	21.25

For day-ahead forecasting (**Table 8**), where the baseline is the physics-based NAM model, all machine learning approaches demonstrate significant value in correcting the NWP output. The deep learning models show a distinct advantage over the linear models. TCN and Crossformer yield excellent results, with GHI skills exceeding 40%. Once again, our proposed model ("Ours*") delivers the best performance by a substantial margin. It achieves a GHI skill of 49.42% and a DNI skill of 37.89%. This demonstrates a powerful ability to learn and correct the complex, non-linear errors inherent in NWP models, establishing a new state-of-the-art performance for day-ahead solar forecasting.

Table 8. Comparative analysis of intra-hour forecasting performance across Ours and competing approaches

Forecasting Horizon	Models	GHI		DNI	
		RMSE (W/m ²)	Skill (%)	RMSE (W/m ²)	Skill (%)
Day-ahead	Baseline	109.72	NaN	245.04	NaN
	OLS	70.07	36.14	196.64	22.93
	Lasso	68.32	37.73	194.16	24.22
	LSTM	67.43	38.54	190.84	25.41
	Informer	66.41	39.48	189.65	26.37
	TCN	65.63	40.18	188.59	26.65
	Crossformer	65.13	40.64	186.58	27.42
	PatchTST	66.77	39.15	183.31	28.97
	Ours*	55.50	49.42	152.19	37.89

6. Conclusion

We have presented a multimodal deep learning framework that demonstrates the critical importance of dynamic data fusion for robust solar forecasting. By intelligently weighing information from sky imagery, satellite data, and meteorological models, our approach achieves state-of-the-art performance, reducing day-ahead forecasting error by nearly 15% compared to other leading architectures and attaining forecast skills of over 49% for GHI and 37% for DNI. This robustness is especially evident in volatile weather where traditional models falter. The key takeaway is that adaptability is paramount; the model's ability to prioritize the most informative data source for any given condition is the primary driver of its success. Future work will focus on enhancing the model's predictive power for extreme ramp events through novel architectural components. We will also investigate transfer learning techniques to improve the model's generalizability to new geographic locations with minimal retraining. Finally, the core adaptive fusion mechanism is a promising approach that could be extended to other complex, multi-modal forecasting problems like wind and load prediction.

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